

Does LIHTC Development Affect Local Housing Costs?

A Bayesian Hierarchical Analysis

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Abstract

Housing policy has become a popular political topic in the U.S., and often revolves around market-rate housing. When affordable housing is addressed it's usually spoken about in a negative light with little empirical evidence. I wanted to understand the relationship between affordable housing, specifically LIHTC, and housing costs (both home and rent prices). I used a Bayesian hierarchical model with county intercepts nested in state intercepts and state-level slopes, and after controlling for population growth and deflating prices using CPI-less-shelter, I find a modest positive association of 0.126% for home values and 0.375% for rents per 1% increase in LIHTC per capita. I used a LOO comparison to compare a few different models and the framework decisively favors the inflation-adjusted hierarchical model. The positive association likely reflects LIHTC allocation patterns rather than a causal price effect, though my analysis indicates that housing policy is too variable across states to have a robust national solution.

1 Introduction

“Affordability” has become a buzzword in the political discourse. Housing affordability has become a massive political talking point, as Americans are overwhelmingly less able to afford a home these days (NAHB, 2024). Many political pundits only discuss public policy surrounding market-rate housing, most notably their desire to lighten local zoning laws to allow for more market-rate development. While I agree with this policy, I notice those pundits often have an aversion to affordable housing programs for a number of reasons. My goal is to understand how affordable housing programs (specifically LIHTC given its scale across the country) affect housing costs, both in the rental market and the housing market.

The Low-Income Housing Tax Credit (LIHTC) is the largest affordable housing creation program in the United States, responsible for over 3.7 million affordable housing units since its creation with the Tax Reform Act of 1986. The federal government (through local housing authorities) allocates $\approx$ \$10.5 billion a year to state and local governments for “the acquisition, rehabilitation, or new construction of rental housing targeted to lower-income households.” (HUD, 2025).

I felt this project was great for a Bayesian approach because I wanted to use a hierarchical model to capture the variation across counties in a state while still recognizing counties within the same state should be considered different from counties in another state. A fully pooled model would ignore counties and states and treat every county-year as a draw from one national population. In reality, counties within a state share similarities, so their housing costs are partially correlated. I also wanted a posterior distribution to help me compare different models I ran on the data.

2 Data

I collected data from five sources to get the number of LIHTC units built in a county, the typical home price and rent price of a county, inflation indices, and county populations.

2.1 LIHTC Database

The Department of Housing and Urban Development (HUD) has a LIHTC database describing over 54k projects from 1987–2023. The projects are geocoded with FIPS codes, which describe a state-county-census tract. This is what I used to connect counties across datasets. I then aggregated so each row is the number of low-income units built per county per year. I had to exclude $\approx 20.9\%$ of projects due to invalid county FIPS codes. HUD masks the location of every New Mexico project, so New Mexico drops out of both datasets.

2.2 Zillow Home Value Index (ZHVI)

Zillow Research creates a Zillow Home Value Index (ZHVI) which represents the “typical” home value (dollar-denominated) for homes in the 33rd–67th percentile range (Zillow, 2025a) for a given geographic level from 2000–present. I downloaded the smoothed and seasonally adjusted ZHVI by county, and annualized the index by taking the mean of monthly values per county-year.

2.3 Zillow Observed Rent Index (ZORI)

Zillow Research also creates a rent index (ZORI) which is a dollar-denominated “typical” market-rate rent for a given geographic level (Zillow, 2025b). I used a similar methodology as above where I took an annual average of smoothed and seasonally adjusted ZORI. ZORI had much spottier coverage: the rental dataset covers 610 counties, compared with 2,073 in the housing dataset.

2.4 County Population Estimates

I wanted to account for housing demand as this would affect increases in both home values and rent prices, so I pulled in Census population data from the U.S. Census Bureau Population Estimates Program: 2000–2009 intercensal estimates (Census Bureau, 2010), 2010–2019 vintage estimates (Census Bureau, 2020), and 2020–2023 estimates via USDA ERS (USDA ERS, 2025).

2.5 Inflation Less Shelter Costs

A major confounding factor in increasing housing costs is inflation, which would always cause prices to increase regardless of LIHTC units constructed, so I pulled CPI values for “All Items Less Shelter” (series CUUR0000SA0L2) from the Federal Reserve Bank of St. Louis (Federal Reserve Bank of St. Louis, 2026), and “annualized” the data by taking the average CPI value for a given year. I didn’t want to include housing prices because this is the very relationship I’m trying to measure, so I needed to control for the increase in costs of everything else (i.e. how can I account for the U.S. dollar losing value).

2.6 Panel Construction

Across these five sources I ended up with two datasets:

- Housing Dataset: 11,481 county-years across 2,073 counties in 49 states and the District of Columbia (2001–2023) to compare LIHTC’s relationship with home values.
- Rental Dataset: 2,516 county-years across 610 counties in 49 states and the District of Columbia (2015–2023) to compare LIHTC’s relationship with rental prices.

In both datasets, I decided to use the cumulative sum of LIHTC units per capita (1k residents) rather than the total units built in a given year. I used a cumulative sum because once these units are built, they’re a part of the housing supply for future years, so any effect these additional units may have will persist further in time than the year the units were built. Secondly, I scaled the sum relative to the county’s population because adding 100 units of housing to a county with 100k in population may look different than 100 units of housing to a county with 1M in population, so I treat these on a relative scale. For simplicity, I count the District of Columbia as a state in the rest of this report.

3 Model Specification

Both of these datasets went through the same set of models throughout this project, and below are the three model types I will compare and discuss in this report. I fit every model in PyMC (Abril-Pla et al., 2023) and computed diagnostics and LOO comparisons with ArviZ (Kumar et al., 2019).

3.1 Pooled Baseline

I have a pooled baseline which contains a single intercept, single slope, and no hierarchy. The formula is as follows:

$$\log(\text{ZHVI}_{c,t}) = \beta_0 + \beta_1 \cdot \log(1 + \text{LIHTC}_{c,t}^*) + \gamma \cdot \text{PopGrowth}_{c,t} + \varepsilon_{c,t} \quad (1)$$

$$\varepsilon_{c,t} \sim \text{Normal}(0, \tau^{-1}) \quad (2)$$

where $\text{LIHTC}_{c,t}^*$ denotes cumulative LIHTC units per 1,000 residents in county c at year t .

For my priors I used: $\beta_0 \sim \text{Normal}(12, \tau = 0.25)$, $\beta_1 \sim \text{Normal}(0, \tau = 0.001)$, $\tau \sim \text{Gamma}(0.001, 0.001)$, and $\gamma \sim \text{Normal}(0, \tau = 0.001)$. I decided on a more informative prior for β_0 since I had a lot of data on the distribution of $\log(\text{ZHVI})$, which centered around 12. The ZORI model followed the same structure but with a prior of $\beta_0 \sim \text{Normal}(7, \tau = 0.25)$. I fit this baseline on both nominal and inflation-adjusted prices.

3.2 Hierarchical Model with Population Control

I have a hierarchical model with county intercepts nested in state intercepts and state-varying LIHTC slopes. The formula is as follows:

$$\log(\text{ZHVI}_{c,t}) = \alpha_c + \beta_{s[c]} \cdot \log(1 + \text{LIHTC}_{c,t}^*) + \gamma \cdot \text{PopGrowth}_{c,t} + \varepsilon_{c,t} \quad (3)$$

$$\alpha_c \sim \text{Normal}(\alpha_{s[c]}^{\text{state}}, \tau_\alpha^{-1}) \quad (4)$$

$$\alpha_s^{\text{state}} \sim \text{Normal}(\mu_\alpha, \tau_{\text{state}}^{-1}) \quad (5)$$

$$\mu_\alpha \sim \text{Normal}(12, \tau = 0.25) \quad (6)$$

$$\tau_{\text{state}} \sim \text{Gamma}(0.001, 0.001) \quad (7)$$

$$\tau_\alpha \sim \text{Gamma}(0.001, 0.001) \quad (8)$$

$$\beta_s \sim \text{Normal}(\mu_\beta, \tau_\beta^{-1}) \quad (9)$$

$$\mu_\beta \sim \text{Normal}(0, \tau = 0.001) \quad (10)$$

$$\tau_\beta \sim \text{Gamma}(0.001, 0.001) \quad (11)$$

I fit this model in non-centered form: the sampler draws standard normal offsets and scales them, for example $\alpha_s^{\text{state}} = \mu_\alpha + z_s \sigma_{\text{state}}$ with $z_s \sim \text{Normal}(0, 1)$ and $\sigma_{\text{state}} = \tau_{\text{state}}^{-1/2}$, and the same for the county intercepts and the state slopes. The priors do not change. I used this form because the centered version sampled poorly: 49 of its 2,180 parameters had $\hat{R}$ above 1.01, and the non-centered version had none. The final models ran four chains of 4,000 draws after 4,000 tuning steps.

Note: the ZORI model uses the same structure but with $\mu_\alpha \sim \text{Normal}(7, \tau = 0.25)$ and $\tau_\alpha \sim \text{Gamma}(2.0, 0.5)$ to address convergence with sparser data.

3.3 Hierarchical Model with Population and Inflation Control

While adding a hierarchical model improved the predictive ability of the model, inflation was a confounding factor as prices would always increase regardless of LIHTC units built, so I decided to take inflation into account. The formula and hierarchical model follow the above, except with index values being calculated as:

$$\text{Index}_{c,t}^{\text{Real}} = \text{Index}_{c,t} / (\text{CPI}_t / \text{CPI}_B)$$

where CPI_B is the CPI of the earliest available year in the housing and rental datasets (2001 and 2015 respectively).

3.4 Revision to the First Version

The first version of this report pooled the county intercepts toward one national mean, $\alpha_c \sim \text{Normal}(\mu_\alpha, \tau_\alpha^{-1})$, while it pooled the LIHTC slopes by state. This mismatch could let part of a high-cost state's price level load onto its LIHTC slope. Nesting the county intercepts in state intercepts fixed the mismatch, and LOO preferred the nested model by 121.63 ± 14.52 ELPD for home values and 104.90 ± 9.98 ELPD for rents (Section 4.3).

4 Results

4.1 Home Values (ZHVI)

Below are the model results for the final ZHVI hierarchical model.

Table 1: Posterior Summary: ZHVI Hierarchical Model (Inflation-Adjusted, Population-Controlled)

Parameter	Mean	SD	2.5% HDI	97.5% HDI	ESS Bulk	ESS Tail	$\hat{R}$
μ_α	11.562	0.049	11.464	11.659	904	1783	1.01
μ_β	0.126	0.022	0.081	0.167	759	1407	1.00
τ_{state}	9.826	2.299	5.555	14.401	1764	3600	1.00
τ_α	9.030	0.314	8.407	9.627	1155	2389	1.01
τ_β	49.783	11.711	27.359	72.565	1988	3760	1.00
γ	1.264	0.146	0.981	1.548	10098	11761	1.00

Across all 2,180 parameters, the largest $\hat{R}$ was 1.008, the smallest bulk ESS was 710, and the sampler had no divergences. Of the 50 state slopes β_s , 32 are credibly positive, 14 have 95% credible intervals that include zero, and 4 are credibly negative: Michigan, Ohio, Illinois, and Connecticut.

Below are the traces for the parameters, the forest plot for the state slopes, and the posterior distribution.

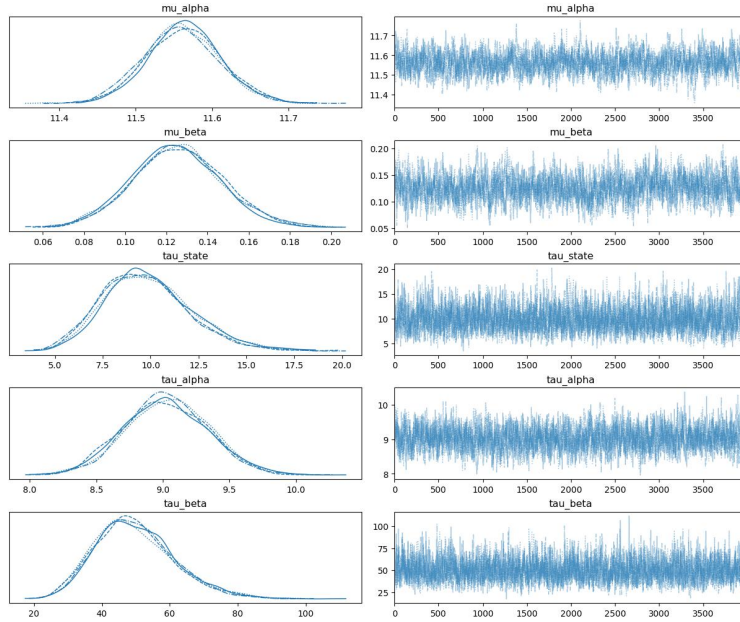


Figure 1: Traces for Bayesian Regression Parameters (ZHVI Model)

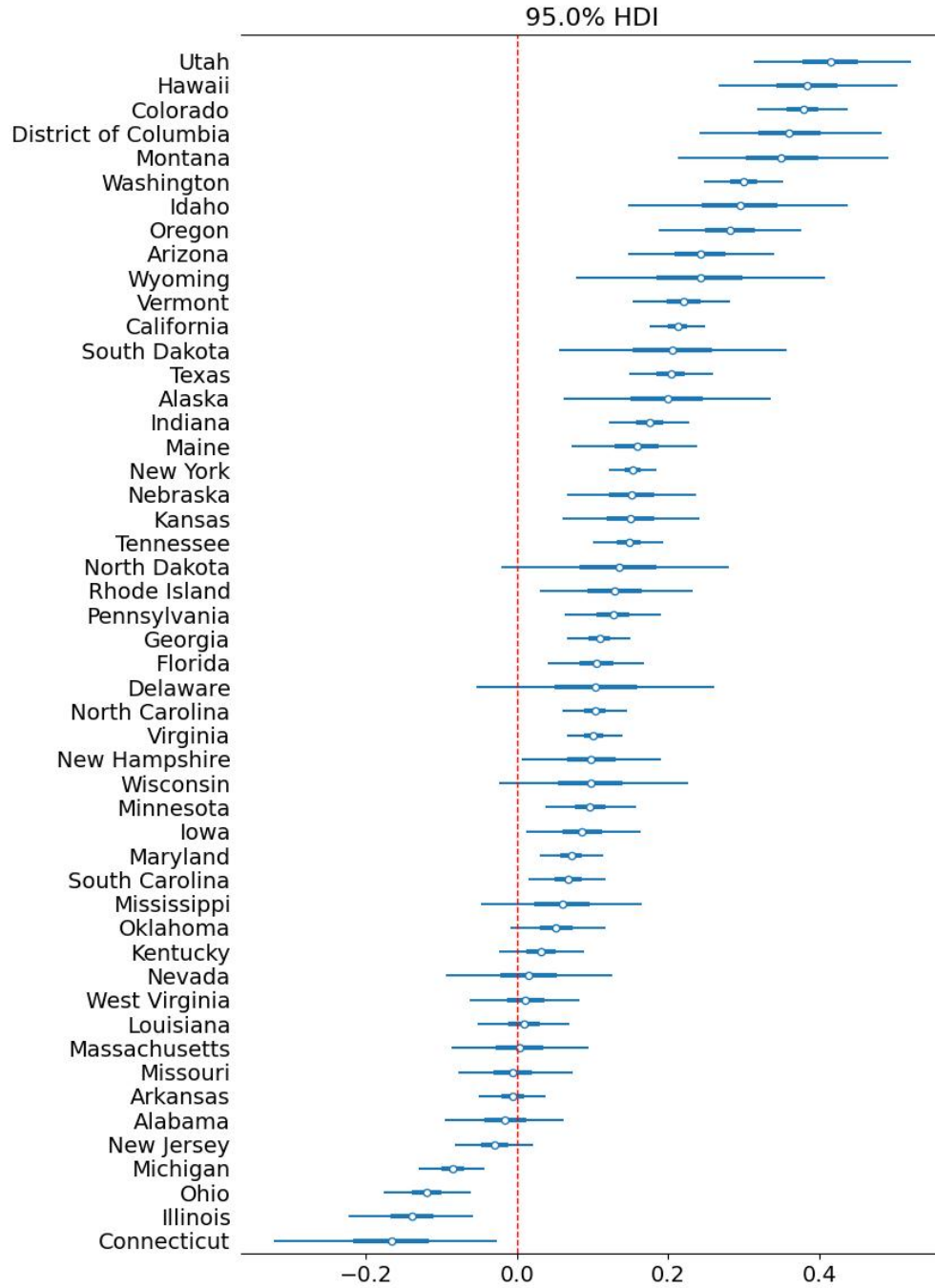


Figure 2: State-level LIHTC slope posteriors (ZHVI model). 95% HDI shown.

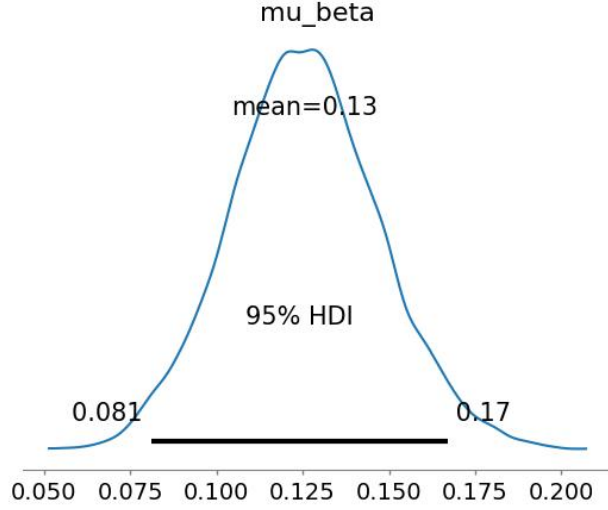


Figure 3: Posterior Distribution of μ_β (ZHVI model). 95% HDI shown.

4.2 Rents (ZORI)

Below are the model results for the final ZORI hierarchical model.

Table 2: Posterior Summary: ZORI Hierarchical Model (Inflation-Adjusted, Population-Controlled)

Parameter	Mean	SD	2.5% HDI	97.5% HDI	ESS Bulk	ESS Tail	$\hat{R}$
μ_α	6.221	0.062	6.100	6.343	15716	11865	1.00
μ_β	0.375	0.030	0.314	0.434	17667	11714	1.00
τ_{state}	10.136	3.469	4.472	17.117	4560	8775	1.00
τ_α	11.284	0.786	9.698	12.780	4188	9410	1.00
τ_β	37.050	12.022	17.069	60.959	4805	8905	1.00
γ	0.077	0.134	-0.186	0.341	18890	11542	1.00

Across all 717 parameters, the largest $\hat{R}$ was 1.002, the smallest bulk ESS was 4,188, and the sampler had no divergences. Of the 50 state slopes β_s , 46 are credibly positive and 4 have 95% credible intervals that include zero: West Virginia, North Dakota, Mississippi, and Louisiana. None are credibly negative.

Below are the traces for the parameters, the forest plot for the state slopes, and the posterior distribution.

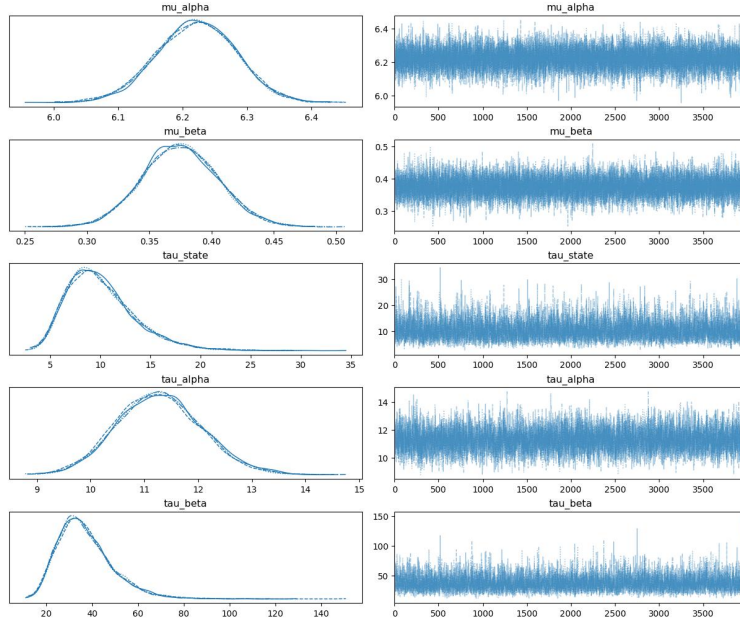


Figure 4: Traces for Bayesian Regression Parameters (ZORI Model)

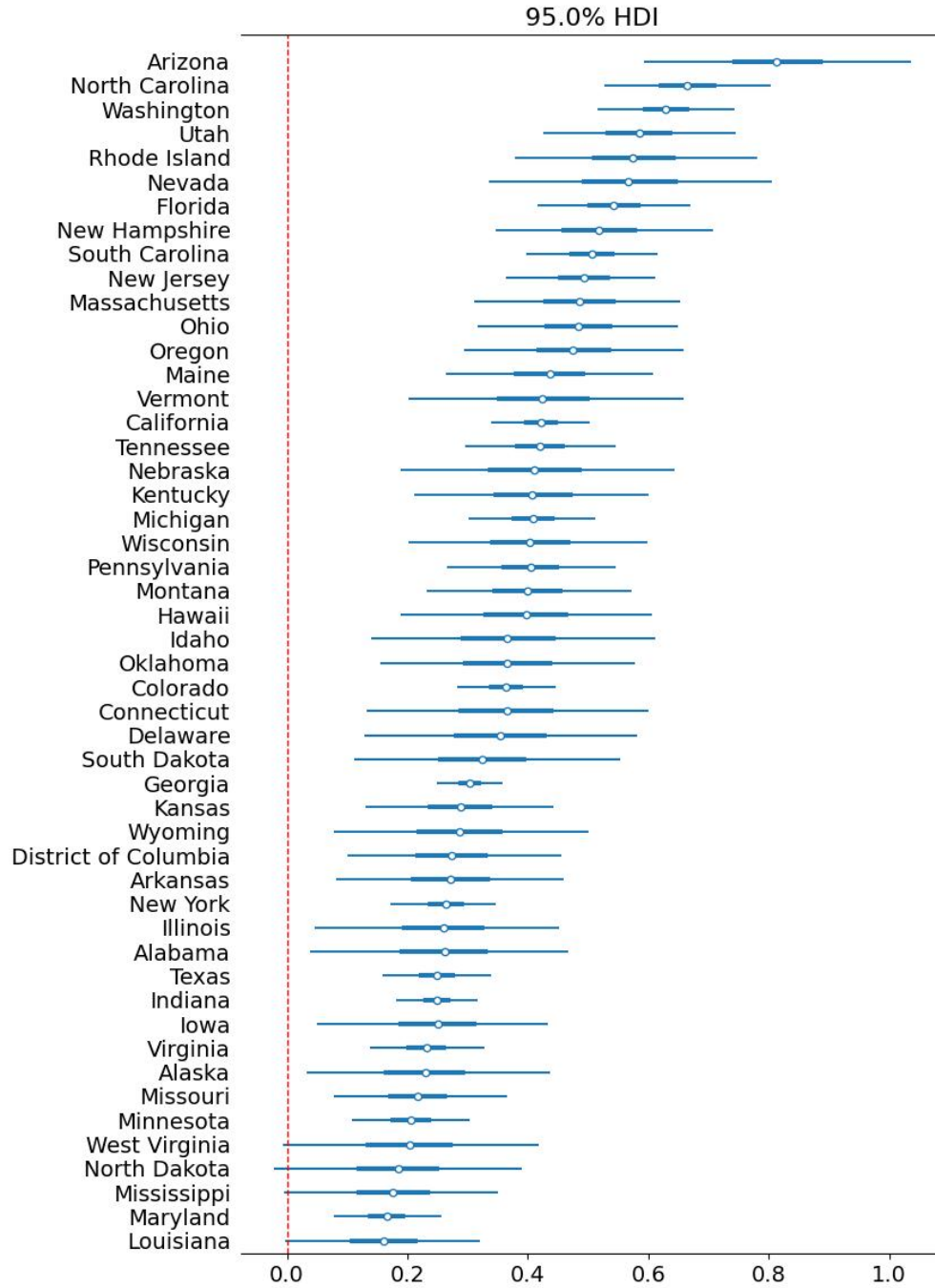


Figure 5: State-level LIHTC slope posteriors (ZORI model). 95% HDI shown.

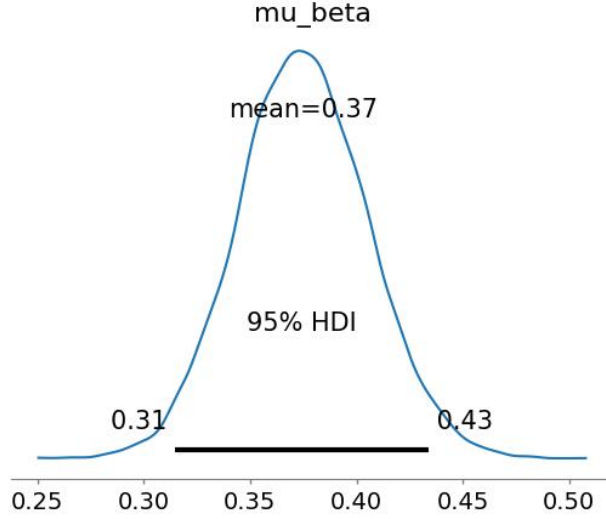


Figure 6: Posterior Distribution of μ_β (ZORI model). 95% HDI shown.

4.3 Model Comparison

For both the ZHVI and ZORI models, I compared their hierarchical model to a pooled version to see the lift in predictive power (i.e. does adding hierarchy help ascertain the relationship between LIHTC units per capita and housing costs). I also compared it with the first version's national intercepts and with the same model on nominal prices. Below are the LOO model comparisons for both ZHVI and ZORI, where Δ ELPD is the gap to the best model and dSE is the standard error of that gap:

Table 3: LOO Model Comparison: ZHVI (Home Values)

Model	ELPD LOO	p_{LOO}	Δ ELPD	dSE
Hierarchical, nested intercepts (Inflation-Adj.)	3407.01	1642.07	0.00	0.00
Hierarchical, national intercepts (Inflation-Adj.)	3285.38	1633.75	121.63	14.52
Hierarchical, nested intercepts (Nominal)	1732.26	1717.80	1674.75	46.06
Pooled (Inflation-Adj.)	-8368.86	13.43	11775.87	122.40
Pooled (Nominal)	-8773.54	11.71	12180.56	117.33

Table 4: LOO Model Comparison: ZORI (Rents)

Model	ELPD LOO	p_{LOO}	Δ ELPD	dSE
Hierarchical, nested intercepts (Inflation-Adj.)	3046.06	590.47	0.00	0.00
Hierarchical, national intercepts (Inflation-Adj.)	2941.16	568.55	104.90	9.98
Hierarchical, nested intercepts (Nominal)	1652.86	588.76	1393.20	24.52
Pooled (Inflation-Adj.)	-671.57	4.71	3717.63	62.01
Pooled (Nominal)	-799.69	4.53	3845.75	59.38

For the final models, the Pareto k diagnostic exceeded 0.7 for 4.34% of the home value observations

and 6.72% of the rent observations, so these LOO estimates are approximate. Every gap in both tables is still more than eight standard errors wide.

5 Discussion

Across both the pooled and hierarchical models, I see a positive association between cumulative LIHTC units per capita and housing costs. Specifically, a 1% increase in cumulative LIHTC units per capita is associated with a 0.126% increase in home prices and a 0.375% increase in rental prices. The effect being stronger for rents than home values makes sense as LIHTC units are rental units, though what is counterintuitive is that somehow adding rental supply is correlated with increasing prices. All else equal, I would expect increasing supply to lower the rent cost. I think this shows there are still several confounding effects, and more so reflects where LIHTC credits are allocated. Housing agencies are probably allocating these credits to areas with demonstrated housing need, which could be more than simply population growth, but things like job growth, wage growth, and zoning changes. For home values, the largest state slopes β_s are in Utah (0.414), Hawaii (0.384), and Colorado (0.378), while Michigan, Ohio, Illinois, and Connecticut are credibly negative. Across states, β_s for home values is correlated with the state's average population growth ($r = 0.505$) and with its average real home value ($r = 0.482$), which indicates that I'm still seeing a relationship tied to population and economic growth, even after controlling for county population growth and scaling LIHTC by population.

While I did not present the result of every model I tested in this report, the ZHVI and ZORI models had much higher β values when working with nominal index values. Under the same nested model, the association between LIHTC units and nominal prices was 0.400% for home values and 0.644% for rents, compared with 0.126% and 0.375% after the inflation adjustment, suggesting a substantial portion of the raw relationship was driven by inflation trends rather than a direct relationship to LIHTC. I also see some of the states' 95% credible ranges contain 0 for the inflation-controlled model when they didn't in the nominal model. For home values, all 50 state slopes were credibly positive with nominal prices; with real prices, 14 include zero and Michigan, Ohio, Illinois, and Connecticut become credibly negative. The states that now include zero range from lower-cost states like West Virginia, Oklahoma, and Arkansas to higher-cost states like Massachusetts and New Jersey, whereas the relationship is highest for states like Utah, Hawaii, and Colorado. For rents, the inflation adjustment moves West Virginia, North Dakota, Mississippi, and Louisiana from credibly positive to including zero.

There were a few limitations on the analysis. One of the bigger data limitations was that ZORI coverage was limited to 2015 onwards, and many counties only had a couple years of data to work with. I think this is where my Bayesian model shined because I didn't need a lot of data to construct my posterior. I used the pooled data to construct my prior which helped the model focus on a specific range of values. Furthermore, the hierarchical structure ensures I could still discern county and state level effects even with limited data. LIHTC data also had missing county FIPS codes for $\approx 20.9\%$ of projects, which may have biased the data towards newer urban projects, but again my Bayesian model was well equipped to handle sparse data.

6 Conclusion

In conclusion, I found that cumulative LIHTC units per capita have a positive association with home prices and rent prices (0.126% and 0.375% respectively) even when trying to account for housing demand (measured in population growth). This association, however, varies greatly by state and shows the benefit of a hierarchical approach, which lets counties in the same state share information because they often share LIHTC allocation rules and housing regulations. The association between housing costs and LIHTC units is higher for rent price, which is sensible since LIHTC units are rental units, and the association is stronger for home values in states like Utah and Colorado where both population growth and housing demand are high, indicating the association is still highly linked to housing demand. Accounting for inflation narrowed the state slopes' 95% credible ranges on average (from 0.184 to 0.167 for home values and from 0.347 to 0.329 for rents) and brought down the strength of the association, indicating much of the nominal relationship was simply linked to inflation. While none of the findings can be considered “causal”, this does suggest housing policy would be best addressed at the state and local level rather than the national level to account for inter-state differences in housing demand and LIHTC effects on housing costs.

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